

Class 09: Candy Mini-Project

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Background

Today we are taking a small detour to analyze a dataset (that we have more intrinsic insight into) with the most useful analysis method we have learned thus far - Principal Component Analysis (PCA)

Data Import

The data is all related to halloween candy and is from the 538 website.

```
candy_file <- "candy-data.csv"

candy = read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0

One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity == 1)
```

```
[1] 38
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```
candy["Nerds", ]$winpercent
```

```
[1] 55.35405
```

Q4. What is the winpercent value for "Kit Kat"?

```
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

There is a useful `skim()` function in the `skimr` package that can help give you a quick overview of a given dataset. Let's install this package and try it on our candy data.

```
# install.packages("skimr")
skimr::skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice-wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes. Most columns are on a 0/1 binary scale. Sugarpercent, pricepercent, and winpercent are on different scales. Winpercent looks especially different because it is a percentage-like variable.

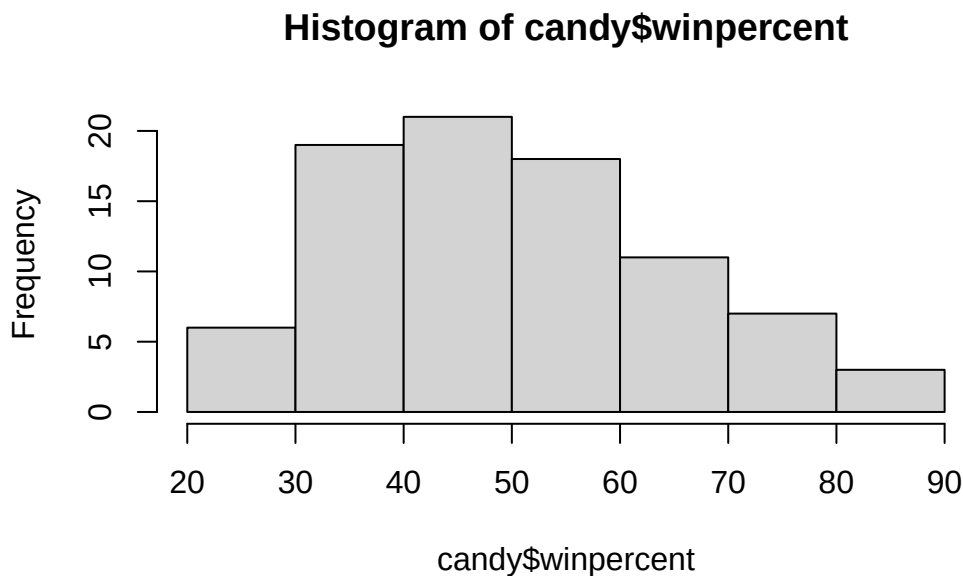
Q7. What do you think a zero and one represent for the candy\$chocolate column?

For candy\$chocolate: 0 = a candy is not chocolate and 1 = a candy is chocolate.

Exploratory analysis

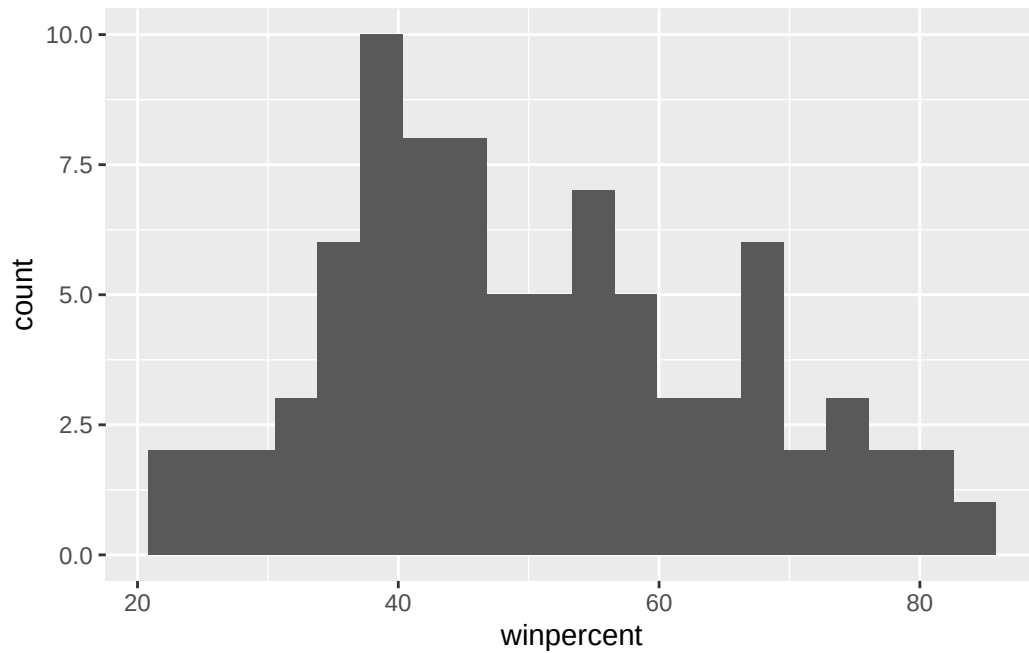
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
hist(candy$winpercent)
```



```
library(ggplot2)

ggplot(candy) +
  aes(winpercent) +
  geom_histogram(bins=20)
```



Q9. Is the distribution of winpercent values symmetrical?

The distribution is not symmetrical.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

```
median(candy$winpercent)
```

```
[1] 47.82975
```

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy$winpercent[candy$chocolate == 1])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[candy$fruity == 1])
```

```
[1] 44.11974
```

On average, chocolate candy is higher ranked than fruit candy.

Q12. Is this difference statistically significant?

```
t.test(
  candy$winpercent[candy$chocolate == 1],
  candy$winpercent[candy$fruity == 1]
)
```

Welch Two Sample t-test

```
data: candy$winpercent[candy$chocolate == 1] and candy$winpercent[candy$fruity == 1]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

The p-value from the Welch Two Sample t-test is 2.871e-08. Since this is much smaller than 0.05, the difference is statistically significant.

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
head(candy[order(candy$sugarpercent, decreasing=TRUE),], n=5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Reese's stuffed with pieces	1	0	0	1	0
Milky Way Simply Caramel	1	0	1	0	0
Sugar Babies	0	0	1	0	0
Skittles original	0	1	0	0	0

Skittles wildberry	0	1	0	0	0
	crispedricewafer	hard	bar	pluribus	sugarpercent
Reese's stuffed with pieces	0	0	0	0	0.988
Milky Way Simply Caramel	0	0	1	0	0.965
Sugar Babies	0	0	0	1	0.965
Skittles original	0	0	0	1	0.941
Skittles wildberry	0	0	0	1	0.941
	pricepercent	winpercent			
Reese's stuffed with pieces	0.651	72.88790			
Milky Way Simply Caramel	0.860	64.35334			
Sugar Babies	0.767	33.43755			
Skittles original	0.220	63.08514			
Skittles wildberry	0.220	55.10370			

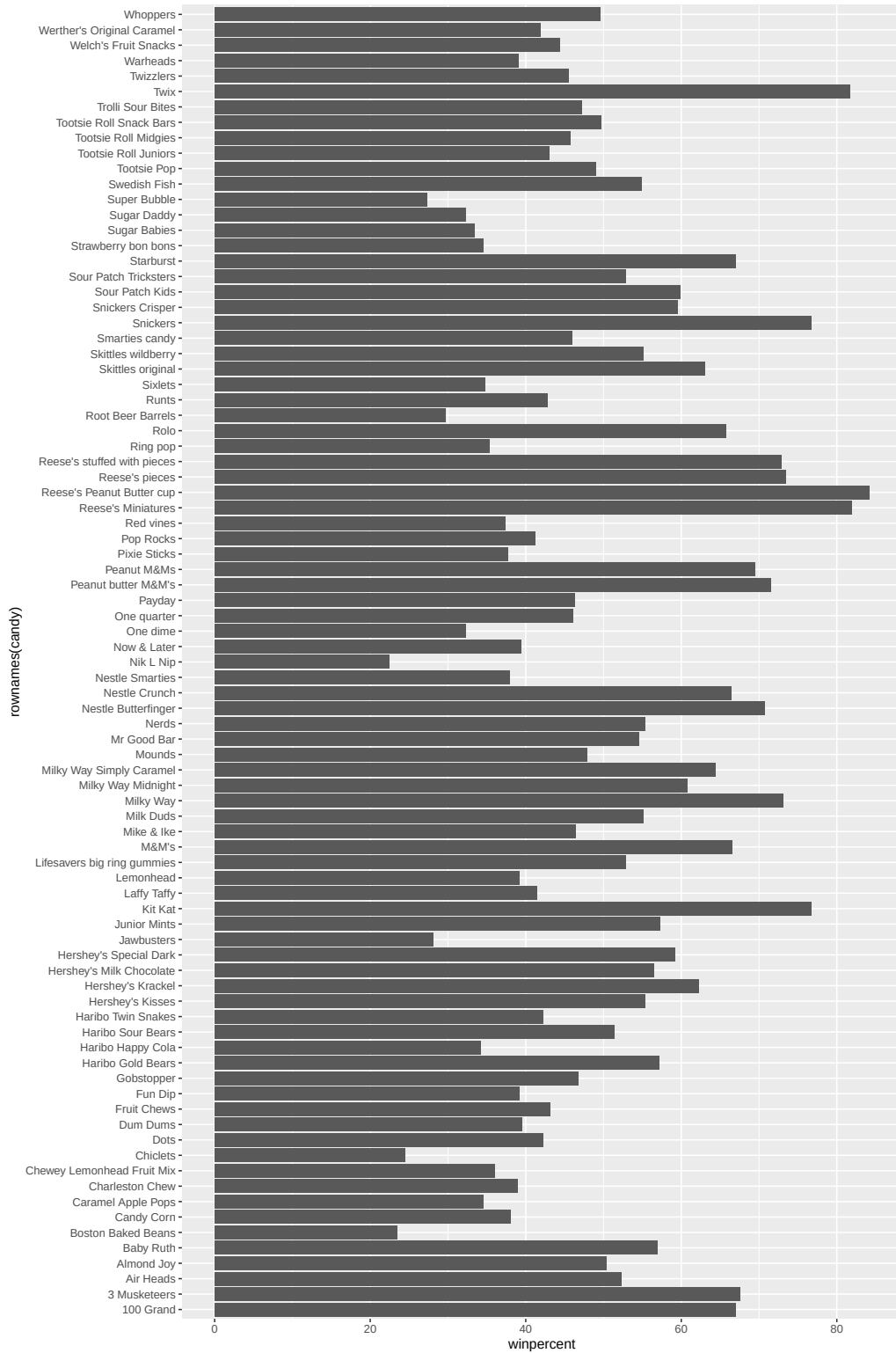
Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy[order(candy$sugarpercent),], n=5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	
One dime	0	0	0	0	0	
One quarter	0	0	0	0	0	
Reese's Miniatures	1	0	0	1	0	
Chiclets	0	1	0	0	0	
Lemonhead	0	1	0	0	0	
	crispedricewafer	hard	bar	pluribus	sugarpercent	pricepercent
One dime	0	0	0	0	0.011	0.116
One quarter	0	0	0	0	0.011	0.511
Reese's Miniatures	0	0	0	0	0.034	0.279
Chiclets	0	0	0	1	0.046	0.325
Lemonhead	0	1	0	0	0.046	0.104
	winpercent					
One dime	32.26109					
One quarter	46.11650					
Reese's Miniatures	81.86626					
Chiclets	24.52499					
Lemonhead	39.14106					

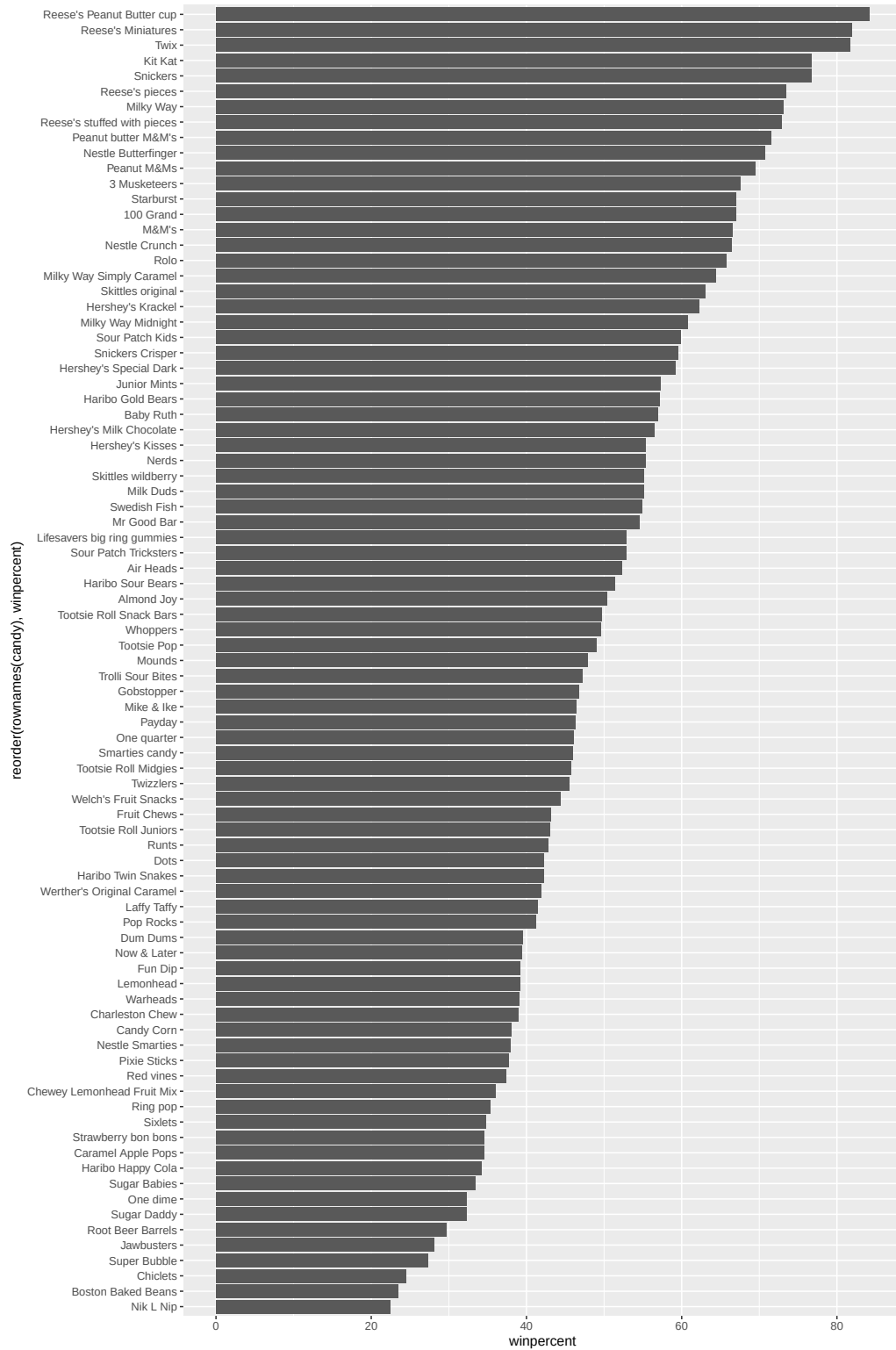
Q15. Make a first barplot of candy ranking based on winpercent values.

```
ggplot(candy) +
  aes(winpercent, rownames(candy)) +
  geom_col()
```



Q16. This is quite ugly, use the `reorder()` function to get the bars sorted by winpercent?

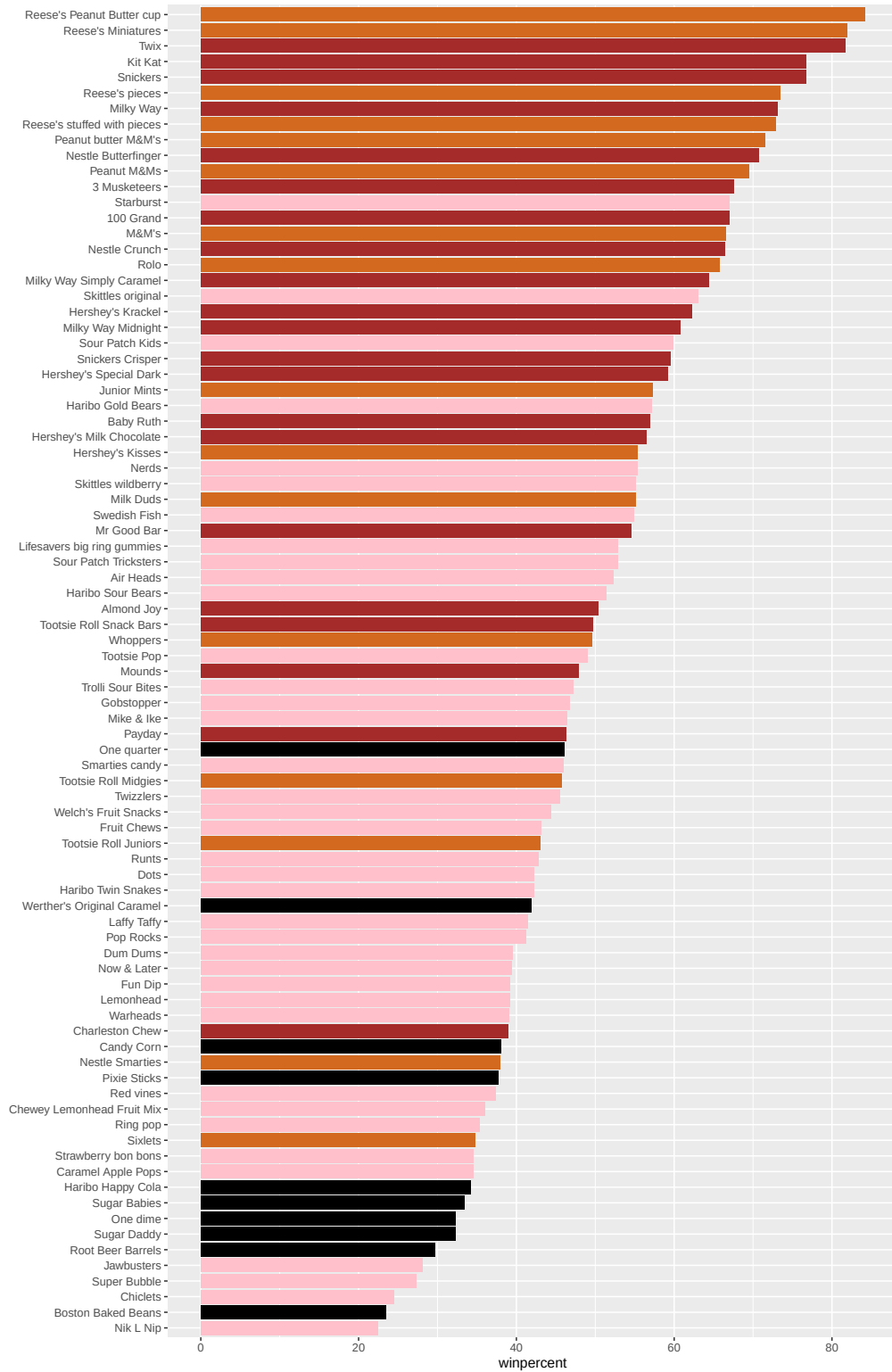
```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy),winpercent)) +  
  geom_col()
```



Q17. What is the worst ranked chocolate candy?

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols) +
  ylab("")
```



The worst ranked chocolate candy is Sixlets.

Q18. What is the best ranked fruity candy?

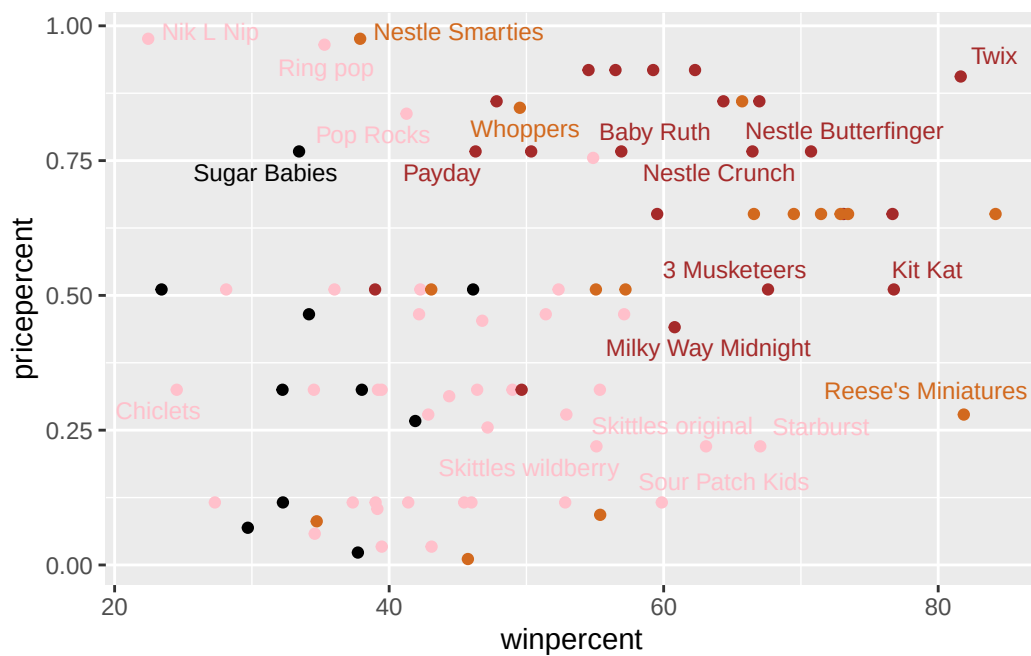
The best ranked fruity candy is Starburst.

Taking a look at pricepercent

We can fix the label text overplotting with an add on package called `ggrepel` and it's `geom_text_repel`.

```
# install.packages("ggrepel")
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(x=winpercent, y=pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

```
ord <- order(candy$pricepercent, decreasing = FALSE)
head( candy[ord,c(11,12)], n=5 )
```

	pricepercent	winpercent
Tootsie Roll Midgies	0.011	45.73675
Pixie Sticks	0.023	37.72234
Dum Dums	0.034	39.46056
Fruit Chews	0.034	43.08892
Strawberry bon bons	0.058	34.57899

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

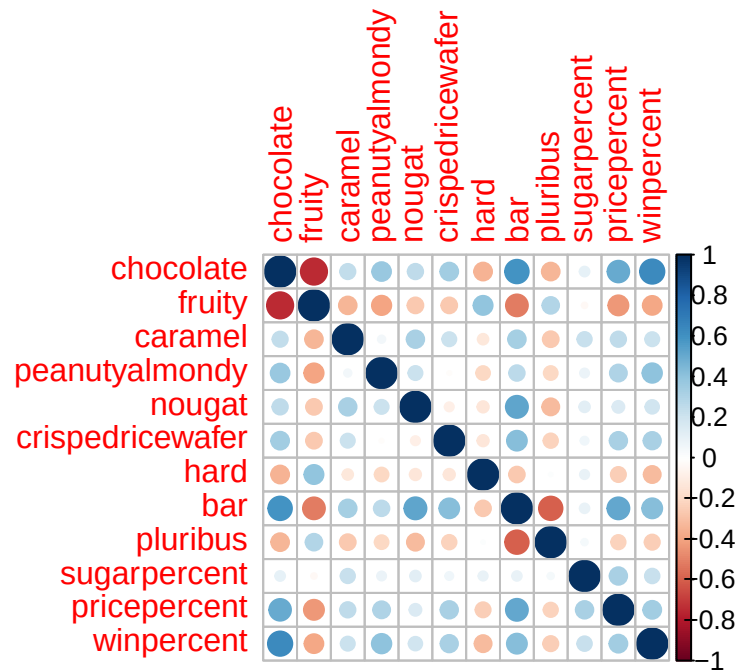
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Exploring the correlation structure

```
# install.packages("corrplot")
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Fruity and chocolate.

Q23. Use your corplot result to make a prediction: which variables do you expect will have the largest contributions (i.e. loadings) to PC1 (i.e., drive the most separation between candies along the first principal component)?

Fruity and chocolate might have the largest contributions (i.e. loadings) to PC1.

Principal Component Analysis

In this case we want to be sure the set `scale=TRUE` because we have one variable `winpercent` that is on a very different scale than all others and would otherwise dominate PCA results.

```
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

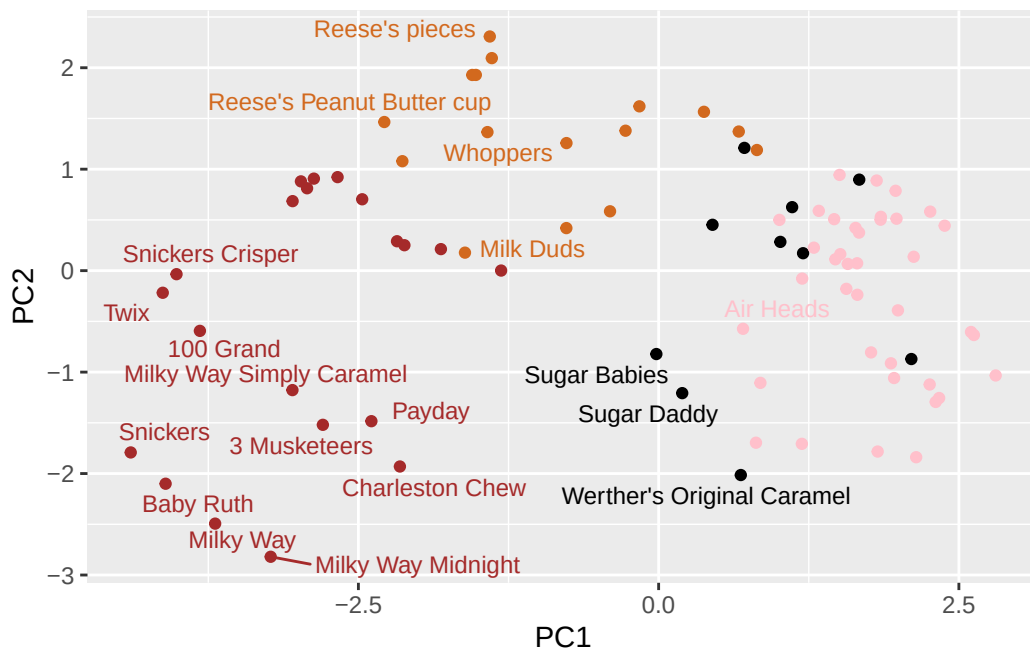
Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539

Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369
	PC8	PC9	PC10	PC11	PC12		
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760		
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317		
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000		

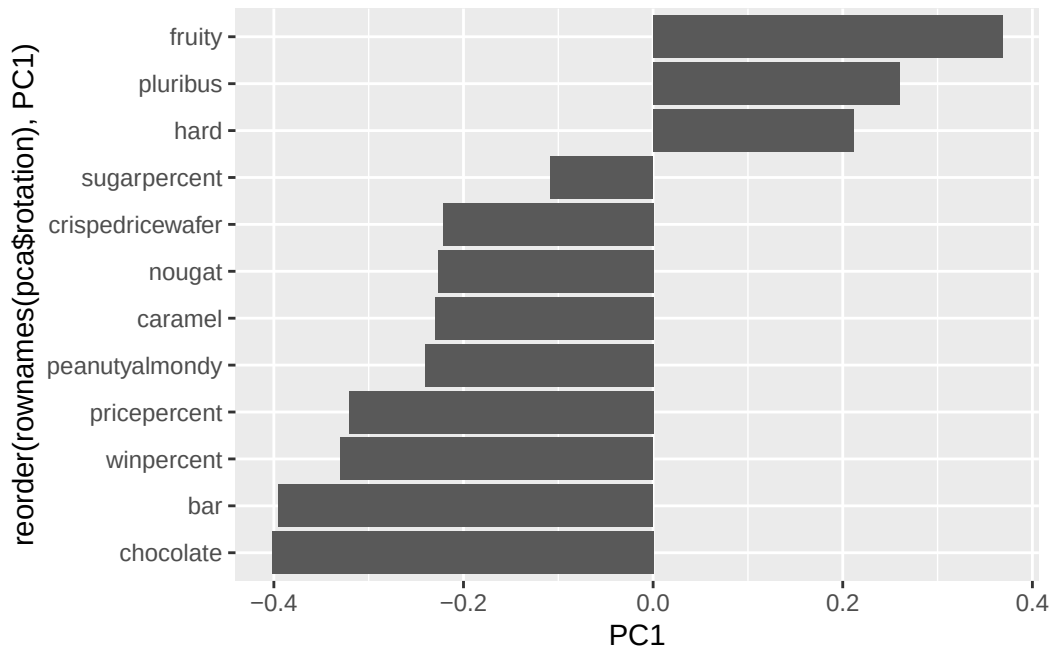
The first major result figure is the “score plot” of PC1 vs PC2 - how different candy are related to each other on our new PC axis:

```
ggplot(pca$x) +
  aes(PC1, PC2, label=row.names(pca$x)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

```
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```

Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

A “winning” candy appears to be one that is chocolate-based, often in bar form, and may include richer ingredients like peanut/almond, caramel, nougat, or crisped rice/wafer. It is less likely to be mainly fruity, hard, or sold as pluribus candy.

The PCA loading plot supports this pattern because chocolate, bar, winpercent, and pricepercent load in the same general direction on PC1, while fruity, hard, and pluribus load in the opposite direction. This suggests that PC1 separates chocolate/bar-style candies from fruity/hard/multiple-piece candies. The correlation plot also supports it as chocolate and fruit are anti-correlated (strong negative correlation coefficients). Other visualization such as ranking of winpercent also indicates that chocolate are more liked.